Medical Image Analysis

BMED365: Topic List B01–B06

BMED365

Spring 2027

Overview

1. MRI Fundamentals
2. Segmentation
3. Quantitative Imaging
4. Tools for Medical Image Analysis

B01: Explain fundamental MRI principles at a high level

Magnetic Resonance Imaging (MRI):

Physical principles (simplified):

1. **Magnetization:** Strong magnetic field aligns hydrogen atoms (protons)
2. **RF pulse:** Radiofrequency pulse "tips" the protons
3. **Relaxation:** Protons return to equilibrium, emitting a signal
4. **Detection:** Signal is measured and reconstructed into an image

Why MRI is important in medicine:

- No ionizing radiation
- Excellent soft tissue contrast
- Many different sequences/contrasts
- Can image function (fMRI)
- Gold standard for brain, joints, spine
- Multiparametric – rich information

Relevance for AI

MRI generates large amounts of data - ideal for AI-based analysis, segmentation, and pattern recognition.

B02: Know about different MR sequences (T1, T1+Gd, T2, FLAIR)

Common MR sequences and their contrast:

Sequence	Characteristics	Clinical use
T1-weighted	CSF dark, fat bright, good anatomy	Anatomy
T1 + Gd	Contrast shows blood-brain barrier breakdown	Active tumor, metastases
T2-weighted	CSF bright, pathology often bright	Edema, inflammation, fluid
FLAIR	T2-like, but CSF dark	Lesions near ventricles, MS plaques
DWI	Diffusion of water molecules	Acute stroke (within minutes!)

For AI analysis

Multiparametric MRI **(multiple sequences)** provides complementary information that AI can leverage for better classification.

B03: Describe segmentation in medical image analysis

Segmentation = identify and delineate structures in images

Task:

- Assign each pixel/voxel to a **class** (e.g. tumor, normal tissue, background)
- Produces a **segmentation map** (mask)

Types of segmentation:

Semantic segmentation:

- Classify each pixel
- All tumors = one class

Instance segmentation:

- Distinguishes individual objects
- Tumor 1, Tumor 2, ...

AI methods for segmentation:

- **U-Net:** Classic encoder-decoder architecture (medical standard)
- **nnU-Net:** Self-configuring, state-of-the-art
- **Segment Anything (SAM):** General-purpose segmentation

B04: Know about the BraTS challenge for brain tumor segmentation

BraTS = Brain Tumor Segmentation Challenge

What is BraTS?

- Annual international competition since 2012
- Benchmark for AI segmentation of brain tumors
- Multiparametric MRI (T1, T1+Gd, T2, FLAIR)

Segmentation task:

- **Whole tumor (WT):** All tumor tissue
- **Tumor core (TC):** Active + necrotic
- **Enhancing tumor (ET):** Contrast-enhancing

Significance:

- Standardized dataset for comparison
- Drives progress in medical AI
- Foundation for clinical translation

Relevance for Team Project

BraTS is directly relevant to the glioblastoma project - students can reference BraTS architectures.

B05: Describe radiomic features and quantitative imaging

Radiomics = quantification of image characteristics

What are radiomic features?

- Mathematical description of texture, shape, intensity in images
- Hundreds of features from a single region of interest (ROI)
- "Deep" properties not visible to the human eye

Feature categories:

Intensity:

- Mean
- Standard deviation
- Histogram features

Texture (GLCM):

- Homogeneity
- Contrast
- Entropy

Shape:

- Volume
- Surface area
- Sphericity

Clinical application

Radiomics can predict molecular markers (IDH status), prognosis, and treatment response - based solely on standard images.

B06: Know about nnU-Net and MONAI as tools

nnU-Net:

- "No-new-Net" – self-configuring
- Analyzes data and selects optimal configuration
- Preprocessing, architecture, training
- State-of-the-art on many benchmarks
- Only requires data – the rest is automatic

Strengths:

- Minimal manual tuning
- Reproducible baseline

MONAI:

- Medical Open Network for AI
- PyTorch-based framework
- Specialized for medical imaging
- Transformations, networks, metrics
- Active developer community

Strengths:

- Flexibility for research
- Pre-built components

Other tools

3D Slicer: Visualization and analysis | **ITK/SimpleITK:** Image processing | **Nipype:** Neuro-pipelines

Summary: Medical Image Analysis

Key points:

- **B01:** MRI principles – magnetization, RF pulse, relaxation, detection
- **B02:** MR sequences – T1, T1+Gd, T2, FLAIR, DWI
- **B03:** Segmentation – classify each pixel, U-Net, nnU-Net
- **B04:** BraTS – benchmark for brain tumor segmentation
- **B05:** Radiomics – quantitative features for prediction
- **B06:** Tools – nnU-Net (automatic), MONAI (flexible)

Team project relevance

Medical image analysis is at the core of the glioblastoma team project. Apply your knowledge of MRI, segmentation, and radiomics.

Lab 2

Hands-on experience with CNNs for image analysis, including medical applications (MR dementia).